

Master Guide: Multi-Variable Proportions in Intermediate Physics

In the real world, physics rarely happens in simple pairs. Most of the laws in your textbook involve three or more variables interacting at once. This is called **Multi-Variable (or Joint) Proportion**.

To understand these complex equations, the secret is to "freeze" one variable (treat it as a constant) to see how the other two "dance" together.

1. The Foundation: Combining Relationships

When one thing depends on several others, we combine their individual rules into one "Super Rule."

The "And" Rule (Joint Direct Proportion)

If a variable A is directly proportional to B , and it is *a*lso directly proportional to C , then A is proportional to their product ($B \times C$).

- **Logic:** $A \propto B$ and $A \propto C \implies A \propto (B \cdot C)$
- **Explanation:** Imagine you are filling a pool. The amount of water (A) depends on how fast the hose flows (B) and how long you leave it on (C). If you double the flow rate and double the time, you get four times the water.

The Combined Rule (Direct + Indirect)

Physics often mixes these. If A increases with B (Direct) but decreases as D gets bigger (Indirect), we put the "increaser" on top and the "decreaser" on the bottom.

- **Logic:** $A \propto B$ and $A \propto \frac{1}{D} \implies A \propto \frac{B}{D}$
- **Equation Form:** $A = k \frac{B}{D}$ (where k is the constant that balances the units).

- **Explanation:** Imagine the brightness of a light on a paper (A). It is direct to the power of the bulb (B) but indirect to your distance from it (D). More power = more light; more distance = less light.

2. Example Case: The "Motion Trio" ($d = v \cdot t$)

This is the perfect example of a multi-variable relationship where Distance (d), Speed (v), and Time (t) interact.

By keeping one variable constant at a time, we can see three different proportions within this single equation:

1. **Freeze Time:** If you and a friend both run for exactly 10 minutes, the person with the higher **speed** (v) will cover more **distance** (d). This is a **Direct Proportion** ($d \propto v$). *Double the speed = double the distance.*
2. **Freeze Speed:** If you both drive at exactly 60 mph, the longer you drive (**time**, t), the more **distance** (d) you cover. This is a **Direct Proportion** ($d \propto t$). *Double the time = double the distance.*
3. **Combined View:** Since d increases with v and d increases with t independently, the total distance depends on their product: $d \propto (v \cdot t)$.
4. **Freeze Distance:** If you both need to travel exactly 10 miles, the person with the higher **speed** (v) will finish in less **time** (t). This is an **Indirect Proportion** ($v \propto 1/t$). *Double the speed = half the time.*

3. Newton's Second Law (Laws of Motion)

This law links Force (F), Mass (m), and Acceleration (a) through the equation $F = m \cdot a$.

- **The Combined Logic:** Force is directly proportional to both mass and acceleration. If you double the mass and double the acceleration, you need **four times** the force ($2 \times 2 = 4$).
- **Freeze Mass (Direct):** $F \propto a$. Kick a ball harder, it speeds up faster.
- **Freeze Force (Indirect):** $a \propto 1/m$. If you push with the same strength, a heavy truck is harder to accelerate than a bicycle.

4. Newton's Law of Universal Gravitation (Gravitation)

The force of gravity depends on the masses of two objects (m_1, m_2) and the distance between them (r):

$$F = G \frac{m_1 m_2}{r^2}.$$

- The Combined Logic: Gravity increases with mass (Direct) but decreases with distance (Indirect).
- Freeze Distance (Direct): $F \propto m$. Bigger planets pull harder.
- Freeze Mass (Indirect Square): $F \propto 1/r^2$. Increasing the Distance (r) rapidly decreases the Force (F).
Double the distance = 1/4 the gravity.

5. Kinetic Energy (Work, Power, and Energy)

Kinetic energy ($K.E.$) depends on Mass (m) and Velocity (v): $K.E. = \frac{1}{2}mv^2$.

- The Combined Logic: $K.E.$ is directly proportional to mass and the *square* of velocity.
- Freeze Velocity (Direct): $K.E. \propto m$. At the same speed, a larger mass has more energy.
- Freeze Mass (Direct Square): $K.E. \propto v^2$. Double the speed = 4 times the energy.

6. Hooke's Law (Mechanical Properties of Solids)

This relates Stress (σ) and Strain (ϵ) using Young's Modulus (Y): $\sigma = Y \cdot \epsilon$.

- Freeze Material (Direct): $\sigma \propto \epsilon$. More pull = more stretch.
- Freeze Strain (Direct): $\sigma \propto Y$. Stiffer material = more force needed for the same stretch.

7. Ideal Gas Laws (Thermodynamics)

The behavior of gases is the ultimate multi-variable example: $P \cdot V = nRT$.

- **Combined Logic:** Pressure and Volume are together on one side (Indirect), while Temperature is on the other (Direct to both).
- **Freeze Pressure (Direct - Charles's Law):** $V \propto T$. Heat air in a balloon, it grows.
- **Freeze Temperature (Indirect - Boyle's Law):** $P \propto 1/V$. Squash air in a pump, the pressure goes up.
- **Freeze Volume (Direct - Gay-Lussac's Law):** $P \propto T$. Heat a sealed tank, the pressure rises.

8. Centripetal Force (Rotational Motion)

To keep an object moving in a circle, force (F) depends on mass (m), velocity (v), and radius (r): $F = \frac{mv^2}{r}$.

- **Combined Logic:** Force increases with mass and speed (Top) but decreases with radius (Bottom).
- **Freeze Mass & Velocity (Indirect):** $F \propto 1/r$. A tighter turn (smaller r) requires more force.
- **Freeze Mass & Radius (Direct Square):** $F \propto v^2$. Double your speed in a curve, and you need four times the force to stay on track.

Summary Cheat Sheet: Multi-Variable Logic

Law / Equation	Relation	Type	Constant ("Frozen")
Motion ($d = vt$)	d vs ($v \cdot t$)	Joint Direct	(None)
Motion ($d = vt$)	v vs t	Indirect	Distance (d)
Newton's 2nd	F vs ($m \cdot a$)	Joint Direct	(None)
Gravity	F vs m	Direct	Distance (r)

Gravity	F vs r	Inverse Square	Masses (m)
Gas Law	P vs V	Indirect	Temperature (T)
Gas Law	V vs T	Direct	Pressure (P)

9. Practice Exercises

Section A: Mixed Variable Logic

1. If $X \propto Y$ and $X \propto Z$, what happens to X if you double Y and triple Z ?
2. If $A \propto \frac{B}{C}$, what happens to A if B is doubled and C is also doubled?
3. In the equation $Q = \frac{k \cdot M}{P}$, if M stays the same but P is cut in half, what happens to Q ?
4. If Y is directly proportional to X^2 , and X increases from 2 to 6 (a 3-fold increase), how many times larger does Y become?
5. A variable Z is joint-proportional to m and h . If m increases by 20% and h decreases by 20%, does Z stay the same? (Hint: $1.2 \times 0.8 = ?$)

Section B: Motion Trio

6. You travel 100 miles at 50 mph. If you want to travel the same 100 miles in half the time, what must your new speed be?
7. Two cars drive for 2 hours. Car A goes 60 mph and Car B goes 30 mph. Compare their distances using a proportion.
8. If distance is held constant, and you triple your speed, what fraction of the original time will the trip take?

Section C: Newton's 2nd Law

9. You push a 10 kg box and it accelerates at 2 m/s^2 . If you push a 20 kg box with the same force, what is the new acceleration?
10. To keep acceleration the same while doubling the mass of a rocket, how much must the engine's force increase?

Section D: Gravity & Energy

11. If the distance between two stars triples, the gravitational force between them becomes how many times weaker?
12. A car is moving at 10 m/s . If it speeds up to 40 m/s (4 times faster), how many times more Kinetic Energy does it have?
13. Two planets have the same mass, but Planet A has double the radius of Planet B. Which planet would have stronger gravity at its surface?

Section E: Gases & Solids

14. A gas is in a sealed metal tank (constant volume). If you double the absolute temperature, what happens to the pressure?
15. You have a piston full of gas. You double the pressure while keeping temperature constant. What is the new volume compared to the old one?
16. According to Hooke's Law, if a 5 kg weight stretches a wire by 1 mm, how much will a 15 kg weight stretch it?
17. If you have two wires of the same material, but Wire A is twice as "stiff" (higher Young's Modulus) as Wire B, which one needs more stress to stretch the same amount?

Section F: Rotational Motion

18. A car takes a turn of radius 50 m. If it takes the same turn at double the speed, how much more centripetal force is needed?
19. To keep the centripetal force the same while doubling the speed, how much would you need to change the radius of the turn?